

Acceleration and variability in the growth of children's early vocabulary

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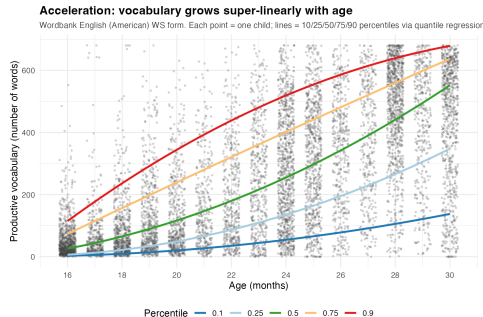
2026

Vocabulary growth: rapid, variable, consequential

Theoretical: Growth mechanisms are diagnostic of how children's learning works — and how it changes with development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has — a *true zero*. Children at birth know zero words. Absolute variability in word counts is interpretable, not just z-scores.

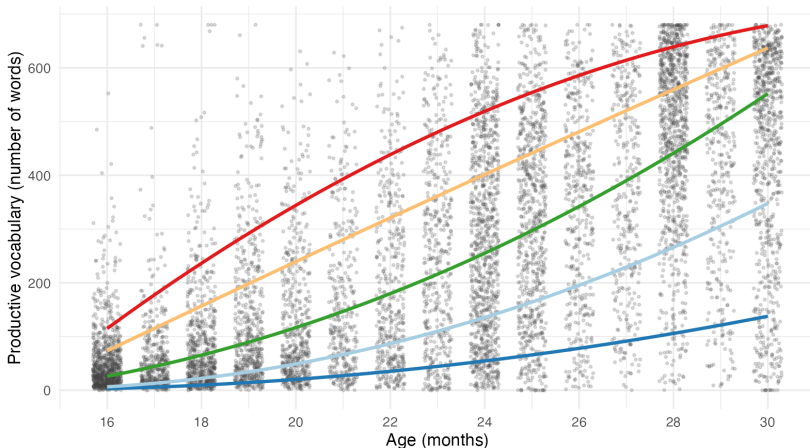


Wordbank English (American) vocabulary vs. age, with quantile fan: rapid growth, large between-child spread, both diagnostic of learning.

Acceleration: vocabulary grows super-linearly with age

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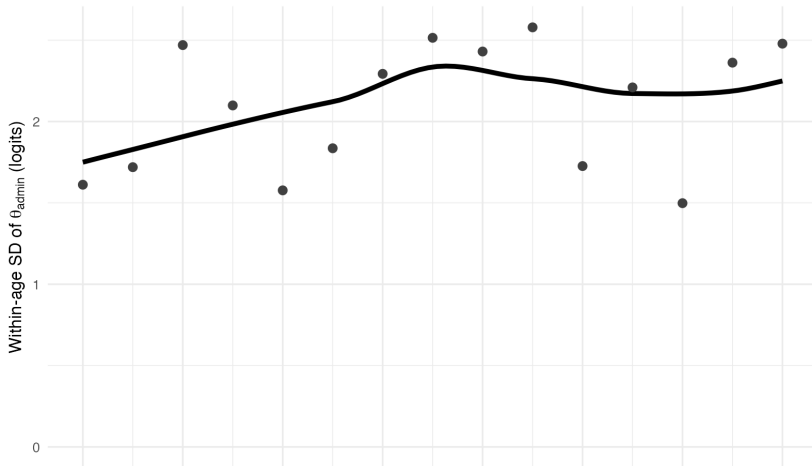
Wordbank English (American) WS form. Each point = one child; lines = 10/25/50/75/90 percentiles via quantile regressor



Variability: between-child spread grows with age

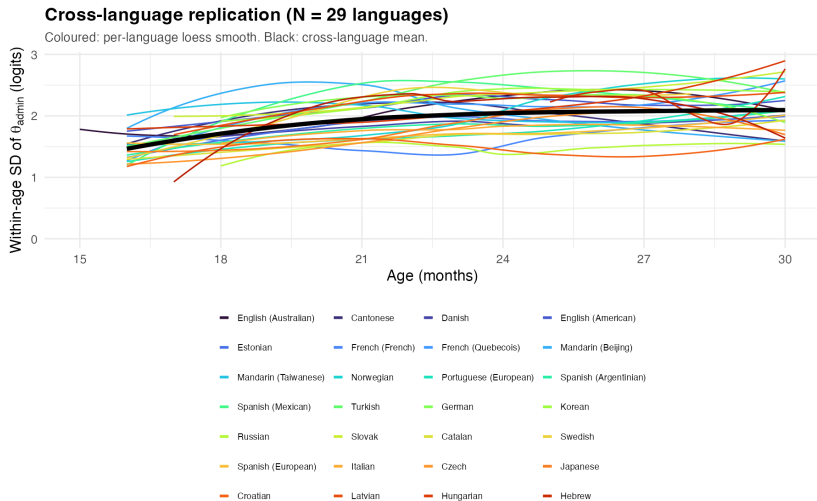
Variability: within-age spread grows with age

Wordbank English (American) WS form. Per-admin theta from 1-PL Rasch IRT.



Cross-language replication

Variability and growth-with-age across Wordbank languages



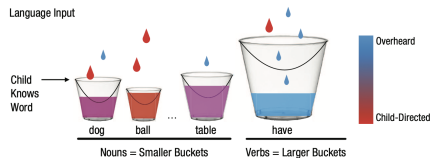
The structural null: pure accumulator

Pure accumulator. Each child accumulates word-evidence at per-child input rate r_i ; word j is acquired when difficulty δ_j is crossed.

Scaling exponent κ_i : ability scales as (tokens) $^{\kappa_i}$.

- $\kappa_i = 1$ for every child: pure unit accumulator.
- Between-child variance comes only from σ_r (input rate).
- No per-child slope variation.

Two estimands diagnose deviation: σ_α (efficiency variation) and the κ_i posterior (is κ above 1, and does it vary?).



Bucket-fill metaphor: each child hears tokens at rate $r_i \cdot p_j$; word j is acquired when its bucket fills past difficulty δ_j (Kachergis et al. 2021).

Prior accumulator models: structural lineage

We'll translate each into our notation later, after the model build

McMurray (2007). Pure accumulator. Each word w has acquisition threshold τ_w from population distribution $g(\tau)$. Acceleration requires $g'(\tau) > 0$.

Mitchell & McMurray (2009). McMurray + leverage feedback $a(t) = t + cL(t)$. Shifts growth without creating acceleration.

Hidaka (2013). Closed-form AoA distributions: cumulative (Gamma), rate-change (Weibull, exponent D on time), or both.

Mollica & Piantadosi (2017). Per-word Gamma waiting time: $\text{AoA} \sim \Gamma(k_w, \lambda_w)$.

Kachergis et al. (2021). 1-PL Rasch IRT with per-child θ_i and linear age covariate.

Each uses different notation. After we introduce our model, we'll re-express each in our coordinates and compare structurally.

The model: Rasch IRT + accumulator dynamics

$$P(\text{produces}_{ij}) = \sigma(\eta_{ij}) \quad \text{with} \quad \eta_{ij} = \theta_i - \delta_j$$

$$\eta_{ij} = \begin{array}{c} \theta_i \\ \uparrow \\ \text{kid ability} \\ \text{(grows with experience)} \end{array} - \begin{array}{c} \delta_j \\ \uparrow \\ \text{word difficulty} \\ \text{(per-word, free)} \end{array}$$

This is a standard Rasch IRT model. The substantive content is in how θ_i depends on accumulated experience. We build that up in three steps.

Step 1: pure accumulator

$$\theta_i = \log r_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = r_i \cdot t$$

$$\theta_i = \log r_i + \log t$$



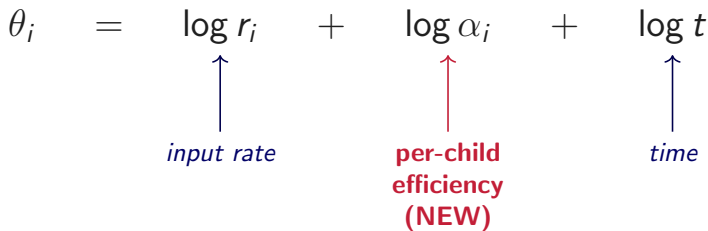
per-child input rate *time = age*
(words per month) *(months since onset)*

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate r_i .

Step 2: + per-child efficiency α_i

$$\theta_i = \log r_i + \log \alpha_i + \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t$$

$$\theta_i = \log r_i + \log \alpha_i + \log t$$



input rate **per-child
efficiency
(NEW)** *time*

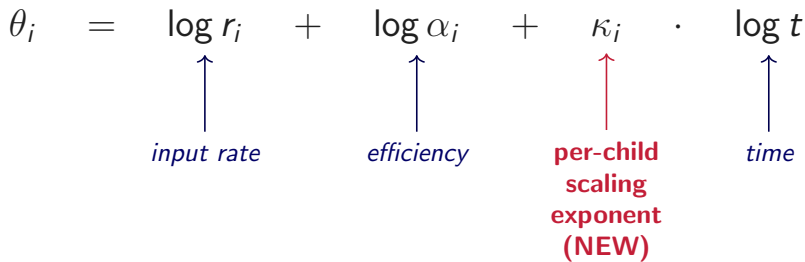
Each child has a multiplicative efficiency α_i over and above their input rate. **Diagnostic**

question: $\pi_\alpha = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2} > 0.5?$ — i.e., is most between-child variation attributable to efficiency rather than input quantity?

Step 3: + per-child scaling exponent κ_i

$$\theta_i = \log r_i + \log \alpha_i + \kappa_i \log t \quad \Longleftrightarrow \quad \exp(\theta_i) = \alpha_i \cdot r_i \cdot t^{\kappa_i}$$

$$\theta_i = \log r_i + \log \alpha_i + \kappa_i \cdot \log t$$


input rate *efficiency* **per-child
scaling
exponent
(NEW)** *time*

Children also differ in how steeply ability scales with cumulative input. **Diagnostic question:** $\kappa_i > 1$? — i.e., is ability super-linear in cumulative tokens, beyond what a unit accumulator predicts?

Prior models in our notation

Each prior accumulator pins specific parameters; ours frees all four

$$\eta_{ij}(t) = \underbrace{\log r_i}_{\text{input rate}} + \underbrace{\log \alpha_i}_{\text{efficiency}} + \underbrace{\kappa_i \log t}_{\text{scaling}} - \underbrace{\delta_j}_{\text{word difficulty}}$$

For each prior model: which of these four parameters are free, which are pinned, and at what values? Walking through five models in the next four slides.

The key distinction we'll see: the prior literature progressively frees parameters — McMurray pins everything; Hidaka frees κ_{pop} ; Kachergis frees σ_α . None frees σ_κ (per-child slope variance). That's what's structurally new in this work.

McMurray (2007) in our notation

The pure accumulator: all four parameters pinned at the unit-rate boundary

Original. Each word w has threshold $\tau_w \sim g(\tau)$; deterministic step at $t \geq \tau_w$. Vocabulary $L(t) = G(t)$ (CDF of g). Acceleration requires $g'(\tau) > 0$ over the productive window.

In our notation:

$$\eta_{ij}(t) = \log t - \log \tau_j = \kappa \log t - \delta_j \quad \text{with} \quad \kappa \equiv 1, \quad \delta_j = \log \tau_j.$$

Parameter	Setting in McMurray
$\log r_i$	$\equiv 0$ (no per-child rate variation)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv 1$ (unit accumulator)
δ_j	free, drawn from g

Hidaka (2013) in our notation

Rate-change form: D is exactly $\kappa - 1$

Original. Three families: cumulative (N -accumulator at constant rate, $\text{AoA} \sim \Gamma$), rate-change (1-accumulator at rate $\delta \cdot t^D$, $\text{AoA} \sim \text{Weibull}$), and both. **Rate-change:** ability $\propto t^{D+1}$; $D > 0$ produces super-linear scaling.

In our notation (rate-change form):

$$\eta_{ij}(t) = (D + 1) \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv D + 1.$$

Parameter	Setting in Hidaka rate-change
$\log r_i$	$\equiv \log \delta_{\text{const}}$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)
κ_i	$\equiv \mathbf{D} + \mathbf{1}$ (shared, no per-child variation)
δ_j	free

Hidaka's $D = \kappa_{\text{pop}} - 1$ in our notation. His $D > 0$ is our $\kappa_{\text{pop}} > 1$.

Mollica & Piantadosi (2017) in our notation

Per-word Gamma waiting time, no per-child structure

Original. Per-word: k_w effective learning instances arrive as Poisson at rate λ_w ; word acquired after k_w events. $AoA_w \sim \Gamma(k_w, \lambda_w)$. For large k_w , Gamma CDF approximates Gaussian (mean k_w/λ_w) and then a ~ 1.7 -scale logistic.

In our notation:

$$\eta_{ij}(t) = \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv 1, \quad \delta_j \approx \log(k_j/\lambda_j).$$

Parameter	Setting in Mollica & Piantadosi
$\log r_i$	$\equiv \log \lambda$ (shared population rate)
$\log \alpha_i$	$\equiv 0$ (no per-child variation)
κ_i	$\equiv \mathbf{1}$ (unit-rate accumulator)
δ_j	$\approx \log(k_j/\lambda_j)$, free per-word

Kachergis et al. (2021) in our notation

First prior model to free σ_α ; σ_κ still pinned

Original. 1-PL Rasch IRT with per-child latent ability:

$\text{logit } P(y_{ij} = 1) = \theta_i + d_j + \beta_a a_i + \beta_t \log(p_j H a_i)$. Per-child θ_i free; d_j per-word free; shared age slope β_a across all kids.

In our notation:

$$\eta_{ij} = (\log r_i + \log \alpha_i) + \kappa \log a_i - \delta_j \quad \text{with} \quad \kappa \equiv \text{shared population slope.}$$

Parameter	Setting in Kachergis 2021
$\log r_i$	free, no per-child variance specified
$\log \alpha_i$	free (their θ_i absorbs $\log r + \log \alpha$)
κ_i	$\equiv$ shared population slope (no per-child variance)
δ_j	free (their d_j)

First prior model to free σ_α . But σ_κ is still zero — every kid has the same scaling slope.

Unified comparison: who pins what

The progression: McMurray → Hidaka → Kachergis → this work

Model	σ_α	κ_{pop}	σ_κ	Per-word δ_j ?
McMurray (2007)	0	1	0	free, $\delta_j = \log \tau_j$
Mitchell & McMurray (2009)	0	1*	0	free, plus leverage c
Hidaka (2013) cumulative	0	1	0	free, $\delta_j = \log(N_j/\delta)$
Hidaka (2013) rate-change	0	$D+1$	0	free
Mollica & Piantadosi (2017)	0	1	0	free, $\delta_j = \log(k_j/\lambda_j)$
Kachergis et al. (2021)	free	free	0	free
Best-fitting model (this work)	free	free	free	free

What's structurally new. Every prior accumulator model has $\sigma_\kappa = 0$ (same scaling slope for all kids). Our model frees it and finds $\sigma_\kappa \approx 3.5$ in English — comparable to σ_α , with $P(\kappa_i > 1) > 0.997$ across kids. *This* is the load-bearing addition that lets the model fit both the acceleration and the variability signature jointly.

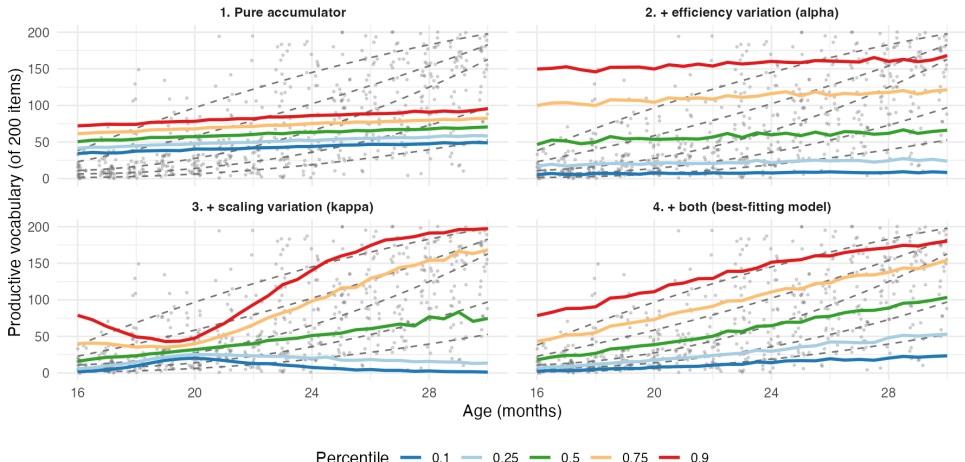
* Mitchell & McMurray's leverage isn't linear in $\log t$; their model lives outside our power-law family.

Each addition matters: 4-panel comparison to data

Variants fit to Wordbank English WS; predicted percentile fans vs. observed

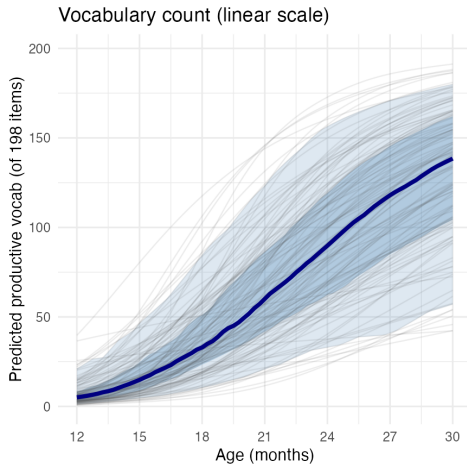
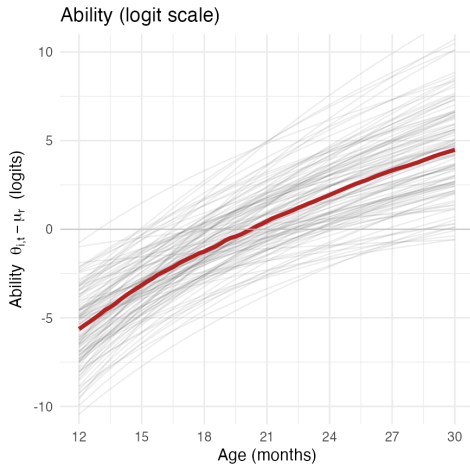
Each addition matters: predicted vs. empirical percentile fans

Empirical (Wordbank): grey points + dashed lines. Predicted (each variant): coloured lines.



Best-fitting model: per-child trajectories

Acceleration and variability both visible



Best-fitting model: parameter posteriors

Acceleration and variability both bounded firmly above zero

Parameter	Posterior	What it does
σ_α	1.71 [1.55, 1.91]	Between-child efficiency variation.
π_α	0.91 [0.89, 0.93]	Share of intercept variance from efficiency, not input.
κ_{pop}	10.35 [10.21, 10.51]	Population scaling exponent: ability scales as (tokens)$^\kappa$.
σ_κ	3.47 [3.14, 3.84]	Between-child variation in scaling exponent.
$\rho(\xi, \kappa)$	+0.35 [+0.22, +0.47]	Higher-baseline kids have steeper growth.

Variability

$\pi_\alpha = 0.91$: efficiency variance dominates input variance.

Acceleration

$\kappa_{\text{pop}} \approx 10.4 \gg 1$: scaling is wildly super-linear, far from the unit accumulator.

Variation in scaling

$\sigma_\kappa > \sigma_\alpha$: kids vary more in scaling exponent than in baseline.

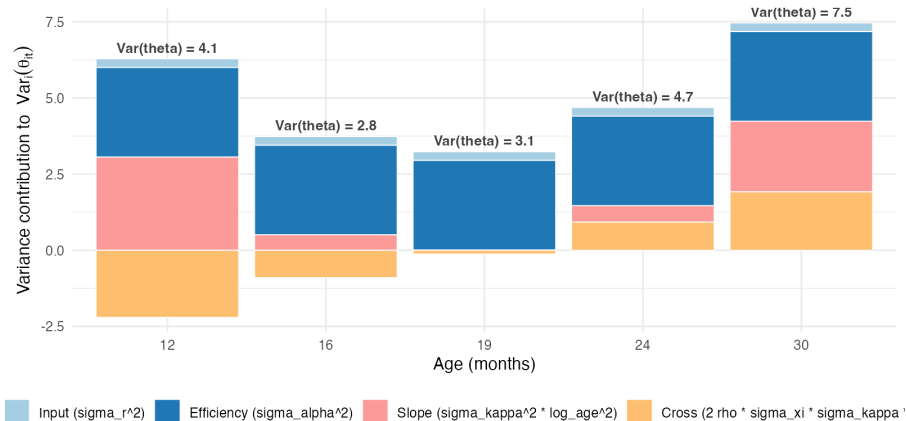
Where between-child variance comes from, by age

π_α is exact at a_0 and qualified elsewhere

Between-child variance composition by age (M_best)

$\sigma_r=0.53$ (input), $\sigma_\alpha=1.71$ (efficiency), $\sigma_k=3.47$ (slope spread), $\rho(x_i,k)=0.35$.

At $a_0=19$ mo, $\log_age=0 \rightarrow$ slope and cross terms vanish; $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2) = 0.91$ exact.



Cross-sample variability: π_α replicates

Five samples, two languages, three input-observation channels

Sample	N	σ_α	π_α
English Wordbank longitudinal	200	1.83 [1.65, 2.04]	0.91 [0.90, 0.93]
Norwegian Wordbank longitudinal	200	2.10 [1.90, 2.35]	0.94 [0.93, 0.95]
BabyView (head-mounted video)	20	1.13 [0.86, 1.61]	0.84 [0.68, 0.93]
SEEDLingS (LENA + comp correction)	44	1.39 [1.13, 1.74]	0.94 [0.89, 0.97]
Peekbank-Stanford (LWL processing)	62	1.64 [1.34, 2.03]	0.90 [0.86, 0.94]

Headline. $\pi_\alpha \in [0.84, 0.94]$ across every well-conditioned sample.

Input rate quantity explains <10% of between-child intercept variance.

Cross-channel replication: directly observed input (BabyView, SEEDLingS), inferred input (Wordbank), processing-coupled (Peekbank). SEEDLingS extreme corrected for parent-report noise via comprehension channel.

Cross-language acceleration: where it lives, not whether it does

Sample	admins/kid	κ_{pop}	σ_{κ}	kid range ($\kappa_{\text{pop}} \pm 2\sigma_{\kappa}$)
English	median 3	10.4 [10.2, 10.5]	3.5 [3.1, 3.8]	~ 3 to 17
Norwegian	median 8	12.4 [12.3, 12.6]	3.7 [3.4, 4.2]	~ 5 to 20

Population scaling exponent $\kappa_{\text{pop}} \sim 10-12$ in both languages. Implied $P(\kappa_i > 1) > 0.997$ in both.

Internal decomposition into δ vs. ζ_i is

identifiability-dependent on longitudinal density.

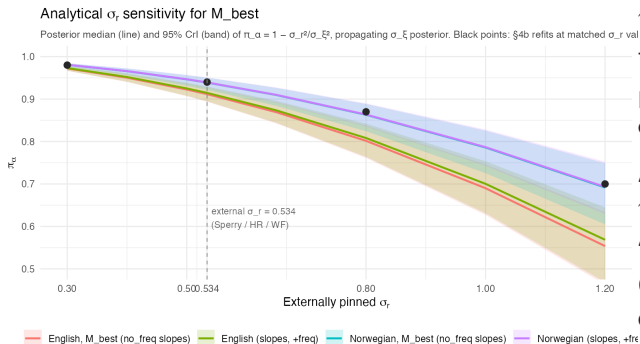
In Norwegian (8 admins/kid) per-child slopes are well-identified, and the population term δ alone — without per-child deviations — hurts fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ .

Reading. The acceleration signature is universal. The parameterization that captures it is identifiability-dependent on data density.

Concretely: in any sufficiently dense longitudinal sample, between-child κ_i spans roughly 5–15.

Stress test 1: how robust is π_α to the input-rate prior?

Analytical sensitivity, validated by refits



$$\pi_\alpha(\sigma_r) = 1 - \sigma_r^2/\sigma_\xi^2.$$

The data identifies σ_ξ^2 tightly; the σ_r prior just *partitions* it into input vs. efficiency.

At external $\sigma_r = 0.534$:

$\pi_\alpha \approx 0.91 - 0.94$.

At doubled $\sigma_r = 1.2$: π_α still > 0.55 .

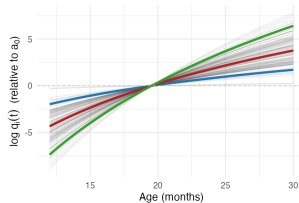
Conclusion. Efficiency-dominated decomposition holds across any defensible external σ_r .

Stress test 2: can input quality absorb acceleration?

Quality-multiplier interpretation: same model, different label

Implied per-child quality multipliers $q_i(t)$

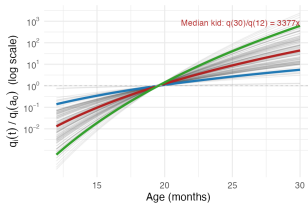
Under the structural equivalence: $\gamma_i = \delta + \kappa_i$, $\gamma_i \sim N(9.35, 3.47^2)$. Quality is a proxy for 'token quality', this is how much quality



kid percentile — 5% — 50% — 95%

Same trajectories on multiplicative scale

Under the structural equivalence: $\gamma_i = \delta + \kappa_i$, $\gamma_i \sim N(9.35, 3.47^2)$. Quality is a proxy for 'token quality', this is how much quality



kid percentile — 5% — 50% — 95%

$\delta + \kappa_i \sim N(9.35, 3.47^2)$. For 'token quality grows with age' to absorb M_best's acceleration, median kid needs $q(30)/q(12) \sim 3377x$; ± 1 SD spread covers 40x to 981046x.

Reviewer reflex. “Maybe kids’ tokens differ in *quality*, and that’s what looks like acceleration.”

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, “quality” would have to grow $\sim 3,000\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

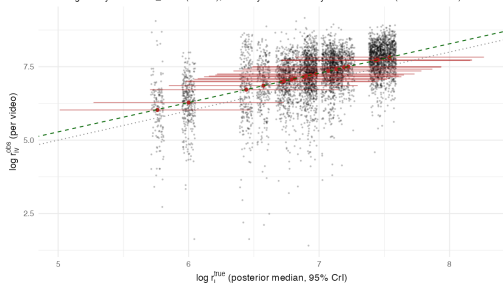
Replicating with directly observed input

BabyView (head-mounted video) and SEEDLingS (LENA all-day audio)

BabyView, $N = 20$

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= 0.28$); dotted: $y = x$. Reactivity inflation = 33% (CrI: -35-173%)

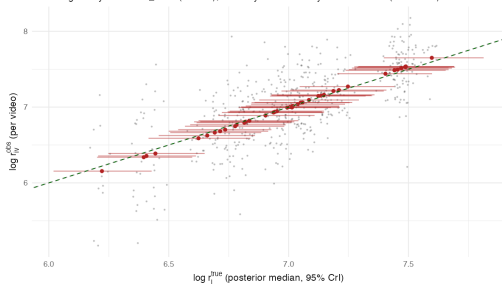


$$\pi_{\alpha} = 0.84 [0.68, 0.93]$$

SEEDLingS, $N = 44$ (with comp correction)

Per-child observed input vs inferred true rate

Dashed green: $y = x + \text{beta_react}$ ($= -0.00$); dotted: $y = x$. Reactivity inflation = -0% (CrI: -0-0%)



$$\pi_{\alpha} = 0.94 [0.89, 0.97]$$

With per-recording observed input rates, the π_{α} headline holds. Input-rate variance is a small

Adding a processing-speed readout: Peekbank LWL channel

$N = 62$ Stanford-linked subjects with both CDI and looking-while-listening

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt} .

Posterior.

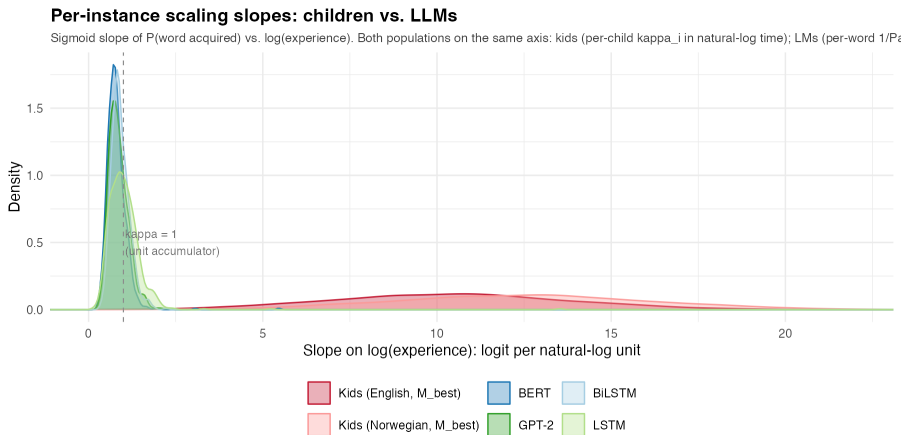
- $\gamma_{rt} = 0.082$ [0.046, 0.125] — LWL processing speed is real, and bounded above zero.
- $\pi_{\alpha} = 0.90$ [0.86, 0.94] — replicates within the Peekbank subset.
- $\rho(\kappa_i, \text{rtslope}_i) = -0.02$ [-0.33, +0.27] — CDI-side and LWL-side growth rates do *not* share variance.

Two clocks, not one. The acceleration in vocab growth and the maturation of processing speed are independent maturation processes within the same child — different “efficiency clocks” ticking in parallel.

This rules out the simplest unification (“one underlying efficiency axis”); the two readouts each pick up a different developmental process.

Direct comparison: per-instance scaling slopes

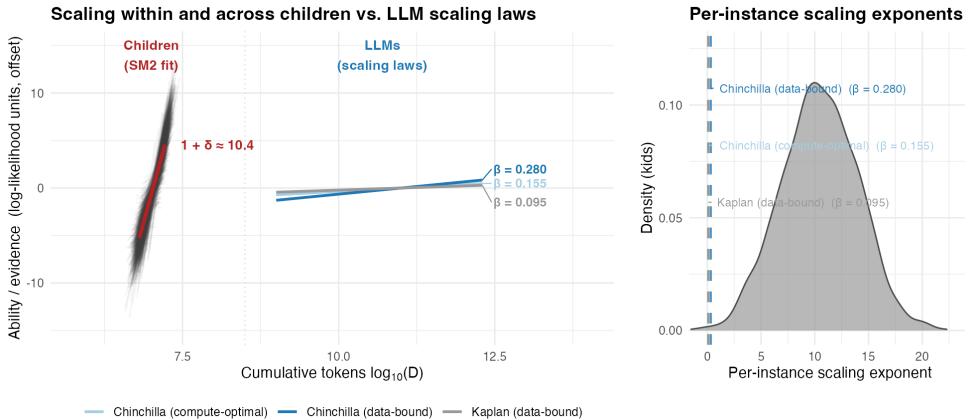
Same task (CDI word acquisition), same axis (logit per log-experience)



Reading. Per-word sigmoid slope on $\log(\text{experience})$, extracted from Chang & Bergen (2022), for 4 LMs $\times \sim 600$ CDI words vs. our per-child κ_i : from the best-fitting model on English and

Coda: scaling regimes are categorically different

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid r_i .



SM2 fit 'long_slopes': median $1 + \delta = 10.39$ [10.24, 10.55]; $\sigma_{\zeta} = 3.48$ [3.15, 3.87]; $\mu_{r_i} = 7.34$, $\sigma_{r_i} = 0.53$, $a_{\zeta} = 19$. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

Mechanisms not in the standard model

What human learning needs that pure accumulators don't have

Sources of acceleration

- Growing processing efficiency with maturation
- Learning-to-learn (mutual exclusivity, shape bias, syntactic bootstrapping)
- Domain-general changes in attention, memory, perception
- Cross-word leverage / vocabulary-mediated bootstrapping

Sources of variability

- Interactional input quality (joint attention, contingent talk)
- Genetic / heritable variation
- Environmental richness not captured by token quantity
- Processing-side individual differences (working memory, attention)

The structural finding constrains theory: any account of early word learning must produce both super-linear-in-tokens growth and large per-child variation in growth rate.

Summary

- Vocabulary uniquely affords *absolute* variability measurement — there is a true zero, and word counts are interpretable in word-count units.
- Two robust signatures across ~ 30 languages and 5 modeling samples: **acceleration** ($\kappa_i \sim 10$, varying widely across kids) and **variability** ($\pi_\alpha \approx 0.91$, growing with age).
- Both signatures violate a pure-accumulator null. Posited “quality” alternatives *relabel* acceleration rather than escape it.
- The picture is structurally different from LLM scaling laws on both axes (steeper exponent, between-instance variance).
- Open question: *which* mechanism — maturation, learning-to-learn, processing speed — is the load-bearing source of the acceleration we measure?